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Homework 2

INST314

1. Identify the control group and the treatment group.

The control group is the individuals given a placebo and the treatment group is the individuals given the drug vaccine.

2. Write a hypothesis for what you think the effect of receiving the Pfizer-BioNTech COVID-19 vaccine was on whether individuals were infected with COVID 19

Individuals who received the Pfizer-BioNTech COVID-19 vaccine are less likely to be infected with COVID-19 compared to those who did not receive the vaccine (placebo).

3. Now test that hypothesis. Select an alpha, choose the correct binomial hypothesis test and run it.

1. HINTS:

1. Separate your data into the treatment and the control groups. (filter function)
 2. Get the sum of the positive COVID tests for each group (use the sum function)
 3. N for each treatment group will be the number of individuals in the groups, not the total number of individuals in the study.
 4. Remember p (the probability of success which in this case is a positive COVID test) *IS NOT* the same thing as p-value (the likelihood of getting a value as or more extreme as the one in your sample if you were to take another sample using the same methodology).
2. Provide me with your hypothesis in words and mathematically, and the results of your hypothesis test both mathematically and in words. H_0 , H_A , alpha etc.) Is this a left-tailed, right-tailed or two tailed test? (do you expect the risk of acquiring COVID to be lower in the test group, higher, or just different, and which would result in evidence to approve the vaccine?)

The results from this study reject the null hypothesis (H_0) since there is a significant difference in the proportion of positive COVID tests between treatment and control groups. The proportion of people testing positive for COVID-19 is different between the treatment (vaccine) and control (placebo) groups. For my significance value (alpha), I chose “.05”. This is a two-tailed test because we are

interested in whether the proportion of positive COVID-19 tests differs between the treatment and control groups in any direction. Since the p-value is less than the chosen significance level (α), we reject the null hypothesis. I expect the risk of acquiring COVID to be lower in the test group since 100% of the vaccinated group has never contracted COVID-19.

4. Was receiving the Pfizer-BioNTech vaccine correlated with a statistically significant reduced risk (lower probability) of getting COVID-19?

Yes, receiving the Pfizer-BioNTech vaccine is directly correlated with a lower probability of getting COVID-19.

5. Does this study make it possible to prove causation (in the sense that any statistically significant difference, if real and not the result of error, is caused by the treatment)? Why or why not?

Yes, we can assume that this study makes it possible to prove causation since 100% of the vaccinated group didn't contract Covid-19. The p-value is also less than our significance value, so we can claim that the number of positive COVID tests differs significantly across the treatment and control groups. While statistical significance suggests an association between receiving the vaccine and a reduced risk of getting COVID-19, it does not necessarily imply causation. Causation cannot be definitively proven without considering the study design and potential confounding factors.

6. Vaccines are intended to trigger a strong immune response so that when a person encounters the pathogen they have been immunized against their body will react quickly and fight the infection off before they can become sick. Triggering a strong immune reaction can carry its own risks. Approval of medications/vaccines etc. relies on weighing the risks to the benefits and determining which is greater. The CDC monitors the safety of newly approved drugs and vaccines. Between approval of the Pfizer-BioNTech vaccine for adolescents on May 10, 2021 and July 16, 2021, approximately 8.9 million US adolescents had received the Pfizer-BioNTech vaccine. The CDC tracking system for adverse events received 9,246 ($p=0.00104$) reports of adverse events- 90.7% ($p=0.00094$) were for mild events such as localized muscle soreness, 9.3% ($p=0.00009662$) were for serious adverse events including allergic reaction and myocarditis (inflammation of the heart) 4.3%. There were 14 reported deaths ($p=0.00000045$): 2 for pulmonary embolism (blood clot that moves to the lungs), two intracranial hemorrhage (bleeding around the brain), 1 for heart failure, two suicides, one for a disseminated *Mycobacterium*

chelonae infection (sepsis) and the rest are unknown or pending. Some of these deaths would be difficult to ascribe to vaccination. Hospitalization rates for adolescents with COVID in the US who have not been vaccinated is 2.1 per 100,000 ($p=0.000021$), with 31.4% ($p=0.000000659$) of those admitted to an ICU. Vaccination confers benefits beyond the individual vaccinated by protecting at-risk individuals around them who might be more severely effected by the infection (herd immunity), but based purely on the above data, which is more risky for adolescents: vaccination Pfizer-BioNTech vaccination, or COVID infection and why?

There are dangers associated with both COVID-19 infection and immunization, although they vary in terms of the severity and likelihood of adverse effects. According to the statistics, major adverse events following the Pfizer-BioNTech immunization appear to be fairly rare in relation to the total number of vaccinations given. Even if the rates of hospitalization and ICU admission for adolescents infected with COVID-19 who have not received the vaccination are low, they still pose a serious danger with the possibility of severe disease and long-term complications. The advantages of vaccination, such as protection against serious disease, hospitalization, and transmission to others, outweigh the dangers for the majority of people, even if vaccination might cause unfavorable outcomes. Most reported occurrences are moderate.

I would argue that vaccination with the Pfizer-BioNTech vaccine appears to pose a lower risk for adolescents compared to COVID-19 infection. However, considerations including age, underlying medical disorders, and local COVID-19 transmission rates should be taken into account in individual risk assessments. Furthermore, continuous study and observation are necessary to assess the safety and effectiveness of vaccines.